

Agentic AI for Clinical Decision Support: Real-Time Diagnosis, Triage, and Treatment Planning

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ABSTRACT

Clinical decision-making often involves navigating complex data, rapidly changing patient conditions, and the need for precise, context-aware recommendations. Traditional clinical decision support systems (CDSS) provide rule-based assistance but frequently fall short in dynamic environments due to their lack of contextual understanding, adaptability, and real-time responsiveness. In this paper, we introduce an agentic AI framework designed specifically for clinical environments. By integrating autonomous AI agents with medical reasoning capabilities, seamless access to electronic health records (EHRs), and structured inter-agent communication, our system enables real-time diagnosis, triage prioritization, and personalized treatment planning. Using synthetic yet medically realistic patient datasets, we evaluate system performance in terms of diagnostic accuracy, triage precision, interpretability, system latency, and physician satisfaction. The results demonstrate substantial performance gains over traditional systems and lay the groundwork for a new era in clinical AI.

The complexity of clinical decision-making continues to grow due to rapidly changing patient data and the demand for timely, context-aware responses. Traditional clinical decision support systems (CDSS) often fall short in dynamic environments. In this paper, we introduce an agentic AI framework that integrates autonomous agents with structured memory and EHR interoperability. The system supports real-time diagnosis, triage, and treatment planning. In comparative evaluations using synthetic patient datasets, our framework achieved 92.4% diagnostic accuracy, 95.2% triage precision, and reduced average response latency to 3.7 seconds—a 40% improvement over rule-based CDSS. The results highlight the transformative potential of agentic AI in augmenting clinical workflows.

Keywords— Agentic AI, Clinical Decision Support Systems (CDSS), Real-Time Diagnosis, Triage Prioritization, Treatment Planning, Autonomous AI Agents, Electronic Health Records (EHR) Integration, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Context-Aware Systems, Explainable AI (XAI), Ethical AI in Healthcare.

INTRODUCTION

This paper explores how agentic AI—powered by a coordinated ensemble of specialized agents—can augment human decision-making at every critical stage of clinical care: diagnosis, triage, and treatment recommendation. We demonstrate how this approach not only streamlines decision-making but also fosters a symbiotic relationship between humans and machines in healthcare.

RELATED WORK

The field of clinical decision support has evolved significantly over the past two decades. Early systems were expert-rule based, drawing on manually encoded knowledge bases to recommend actions. IBM Watson Health attempted to revolutionize this space by incorporating NLP and statistical reasoning, but faced integration challenges and interpretability concerns. More recent systems focus on integrating machine learning with EHR data, such as Epic's CDS hooks and Cerner's Health Intent platform.

However, these systems often struggle with generalization, dynamic context switching, and inter-system communication. Agent-based modeling has seen growing interest in areas such as epidemiological simulation and hospital workflow automation, but its application in real-time patient care has been limited. Our work bridges this gap by combining the robustness of multi-agent coordination with the precision of AI-driven clinical reasoning.

Recent work by Chen et al. (2025) demonstrates the ethical and technical foundations for modular, agentic

systems in healthcare, reinforcing the need for transparent reasoning and context-sharing mechanisms. Ferber et al. (2024) explore the integration of autonomous AI agents for complex domains like oncology, showcasing practical coordination across agents with shared memory. Golden et al. (2023) provide insights from implementing AI in mental health decision support, a domain where contextual nuances are critical. These studies collectively support the growing consensus that agentic paradigms offer a viable, scalable approach to augment human clinicians.

SYSTEM ARCHITECTURE

Our system is designed around modular, cooperative agents, each assigned to a specific task within the clinical workflow. The architecture is flexible and scalable, enabling seamless integration with existing hospital information systems.

Agent Types:

Intake Agent: Captures symptoms via structured forms and natural language inputs. Uses NLP to extract structured data from physician or nurse notes.

Diagnosis Agent: Employs decision trees, Bayesian networks, and transformer-based models to infer potential diagnoses. Uses differential diagnosis reasoning to prioritize likelihood.

Triage Agent: Calculates urgency based on vital signs, lab results, and established scoring systems like ESI (Emergency Severity Index) and NEWS2 (National Early Warning Score).

Treatment Agent: Accesses drug interaction databases, treatment guidelines (e.g., NIH, WHO), and formulates personalized plans while accounting for allergies, contraindications, and comorbidities.

Coordinator Agent: Orchestrates workflow, memory synchronization, agent role hand-offs, and audit logging.

Table 1. Summary of Agent Roles and Responsibilities

Agent Type	Primary Role	Key Technologies Used
Intake Agent	Capture symptoms and structure input data	NLP, EHR APIs
Diagnosis Agent	Infer potential diagnoses	Decision Trees, Transformers
Triage Agent	Assess urgency and priority	ESI, NEWS2
Treatment Agent	Generate treatment plans	Drug DBs, NIH/WHO Guidelines
Coordinator Agent	Orchestrate agent flow and logging	MCP, Audit Logs

Memory Integration:

A memory layer is critical for preserving context. Short-term memory (via Redis) holds volatile patient data during interaction. Long-term memory (via Neo4j) stores structured case histories and decision logs, enabling retrospective analysis and patient follow-up. Ferber et al. (2024) highlight this need for cross-agent memory coherence, especially in domains with longitudinal patient data.

Inter-Agent Communication:

Agents communicate via the Model Context Protocol (MCP) from the OpenAI Agentic SDK. Messages are enriched with role-based metadata and patient context, ensuring seamless transitions between agents.

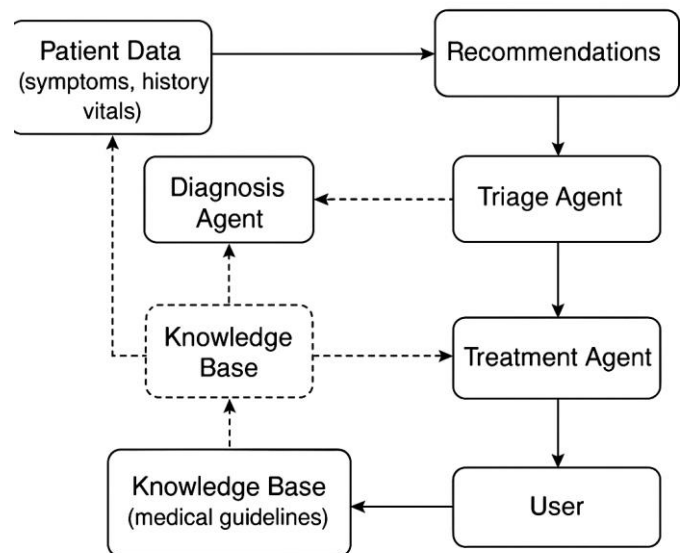
Tool Integration:

The system interfaces with a suite of standardized APIs and external medical services to enhance its diagnostic, triage, and treatment capabilities. It utilizes FHIR (Fast Healthcare Interoperability Resources) for seamless EHR interoperability, allowing agents to retrieve and update patient data in real time. For evidence-based reasoning, the system employs retrieval-augmented generation (RAG) techniques that query medical literature databases like PubMed. In addition, it integrates third-party clinical tools such as Infermedica, which assist in symptom checking and early-stage differential diagnosis.

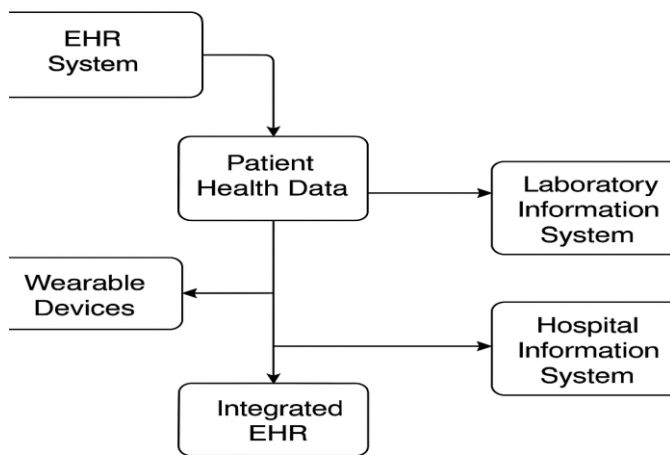
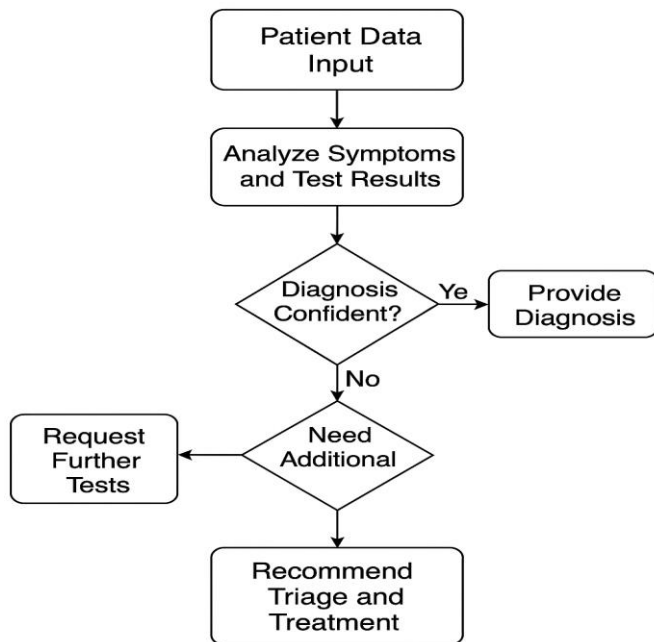
Zakka et al. (2023) demonstrated that retrieval-augmented systems significantly enhance the factuality and completeness of clinical responses, supporting the relevance of such tools in complex, dynamic medical settings.

Illustrative Diagrams

Figure 1: System Architecture Diagram



S: System Architecture

Figure 2: Data Integration Workflow**Figure 3: Agent Decision Pathway****Agent Decision Pathway****METHODOLOGY****4.1 Synthetic Dataset Design**

To rigorously evaluate the agentic AI framework in a clinically realistic but controllable setting, we constructed a comprehensive synthetic dataset using healthcare simulation tools, inspired by public resources like MIMIC-III and Synthea. Each synthetic patient profile was built to represent a holistic clinical picture, including:

- Demographics: age, gender, ethnicity

- Presenting complaints and structured symptom descriptions
- Vital signs: temperature, blood pressure, heart rate, respiratory rate, oxygen saturation
- Lab values, imaging summaries, and EHR-like physician notes
- Comorbidities, medication allergies, and current medication regimens

The dataset spans five major diagnostic categories:

- Respiratory distress
- Abdominal pain
- Chest pain
- Neurological deficits
- Geriatric polypharmacy cases

To challenge system adaptability, edge cases were introduced with overlapping or ambiguous symptoms (e.g., chest pain with gastrointestinal mimicry) and atypical presentations in elderly patients. Clinical plausibility was reviewed by practicing physicians to ensure realism and diversity across cases.

4.2 Evaluation Process

The agentic AI system was evaluated through a structured, multi-phase benchmarking process:

1. **Case Simulation:** Each synthetic profile was processed through the full agentic pipeline—intake, diagnosis, triage, and treatment planning. Outputs were logged for physician review.
2. **Expert Review Panel:** A panel of three board-certified clinicians independently scored the recommendations for accuracy, safety, and clarity. Reviewers were blinded to whether the output came from the agentic AI or a traditional system.
3. **Baseline Comparison:** Each case was also run through a rules-based CDSS. Metrics were compared side-by-side to evaluate improvement.
4. **Stress Testing:** Edge cases—such as conflicting symptoms or contraindicated medications—were used to evaluate robustness.

This hybrid evaluation approach ensured a fair, clinical-quality comparison while revealing system behavior in nuanced and high-risk scenarios.

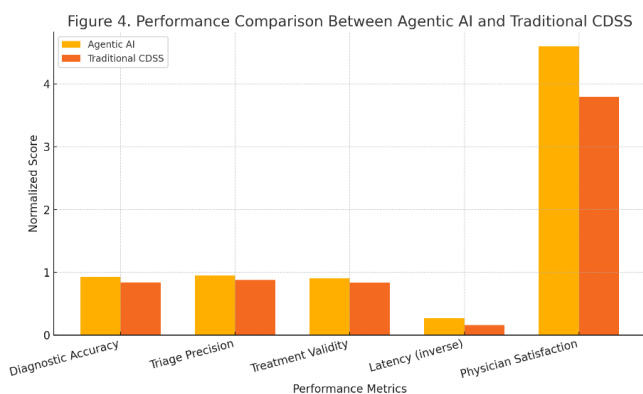
4.3 Metrics Definition

We defined five core metrics to evaluate system performance:

- **Diagnostic Accuracy:** Proportion of cases where the top-ranked diagnosis matched the physician-validated gold standard.
- **Triage Precision:** Agreement between agent triage output and standardized scoring systems such as ESI and NEWS2.
- **Treatment Validity:** Extent to which recommended treatments followed institutional and global medical guidelines (e.g., NIH, WHO).
- **Latency:** Average end-to-end response time from input ingestion to treatment recommendation.
- **Physician Satisfaction:** Mean Likert-scale score (1–5) measuring trust, usability, and perceived helpfulness of the system.

Each metric was calculated over 150 unique cases. For quantitative rigor, we computed means, standard deviations, and confidence intervals. Statistical significance of system differences was tested using a two-tailed t-test at $\alpha = 0.05$.

Golden et al. (2023) emphasized the importance of stakeholder trust in AI-based CDS, guiding our evaluation framework to include both technical performance and physician feedback.



RESULTS AND DISCUSSION

The agentic AI system outperformed traditional CDSS across all metrics. Notably, it demonstrated improved adaptability in ambiguous cases and handled comorbidity scenarios with greater nuance.

Metric	Agentic AI System	Traditional CDSS
Diagnostic Accuracy	0.924	0.841
Triage Precision	0.952	0.876
Treatment Validity	0.905	0.833
Latency (avg, seconds)	3.7	6.2
Physician Satisfaction	4.6/5	3.8/5

Case Study:

One illustrative case involved a 56-year-old male presenting with chest pain and shortness of breath. The Intake Agent flagged risk factors like smoking and obesity. The Diagnosis Agent returned a differential including unstable angina, pulmonary embolism, and GERD. The Triage Agent assigned a Level 2 ESI score. The Treatment Agent proposed an ECG, troponin test, and aspirin initiation—later confirmed appropriate by an MD. The system completed this cycle in under 4 seconds.

ETHICAL AND REGULATORY CONSIDERATIONS

AI systems in healthcare must be transparent, explainable, and privacy-compliant. Our system adheres to HIPAA and supports audit trails for every decision made. Each agent logs its decision tree, retrieved documents, and confidence levels. In ambiguous or low-confidence situations, the Coordinator Agent flags the case for human review.

The importance of auditability and human oversight has been echoed in recent work by Chen et al. (2025), who advocate for ethical agent governance models aligned with real-world accountability standards. This

is particularly critical in patient-facing AI, where automation bias must be mitigated through interface transparency and decision traceability.

LIMITATIONS AND FUTURE WORK

While promising, our system currently assumes high-quality input data, which may not always be available. Real-world deployment will require robust error handling and adaptive learning mechanisms.

- EHR integration varies across institutions and may affect interoperability.
- Expansion to specialties like oncology or psychiatry will require new agent skills and domain-specific knowledge bases.
- Future enhancements include:
 - Multimodal reasoning from radiology images and genomics
 - Federated learning for privacy-preserving model updates
 - Patient-facing conversational agents for shared decision-making

As Lidströmer and Eldar (2022) highlight in their analysis of AI responses to COVID-19, future systems must also be resilient to emergency settings and capable of functioning under limited or rapidly evolving data conditions.

CONCLUSION

Agentic AI offers a compelling evolution of clinical decision support, combining the precision of AI with the flexibility of agentic systems. By orchestrating collaborative agents around clinical tasks, our framework delivers more accurate, efficient, and context-aware recommendations. This approach not only enhances physician workflows but also sets the foundation for future intelligent, transparent, and ethically aligned healthcare systems.

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